

topic_label	topic_words	tax_large	tax_large_sim	ver_large	ver_large_sim	ver_base	ver_base_sim	tax_base	tax_base_sim
Named Entity Recognition	entity, recognition, linking, extraction, named	Named Entity Recognition	1.00	Named Entity Recognition	1.00	Entity Recognition and Recognition	0.77	Entity Recognition Techniques	0.75
Cloud and Edge Computing	cloud, edge, computing, decentralized, processing	Cloud and Edge Computing	1.00	Cloud Edge Computing	0.90	Cloud Computing and Edge Computing	0.97	Cloud Computing and Edge Computing	0.97
Natural Language Processing	language, natural, processing, translation, sentiment	Natural Language Processing	1.00	Natural Language Processing	1.00	Natural Language Processing	1.00	Natural Language Processing	1.00
Natural Language Processing	natural, language, text, sentiment, nlp	Natural Language Processing	1.00	Natural Language Processing	1.00	Natural Language Processing	1.00	Natural Language Processing	1.00
Real-Time Stream Processing	stream, processing, real-time, kafka, flink	Real-Time Stream Processing	1.00	Real-Time Stream Processing	1.00	Real-Time Communication	0.61	Real-Time Stream Processing	1.00
Time Series Forecasting	time, series, forecasting, trending, seasonal	Time Series Forecasting	1.00	Time Series Forecasting	1.00	Time Series Forecasting	1.00	Time Series Forecasting	1.00
Logic Synthesis and Optimization	synthesis, logic, rtl, timing, optimization	Logic Synthesis and Optimization	1.00	Logic-Based Synthesis	0.85	Synthesis and Optimization	0.80	Synthesis and Logic	0.79
Density Functional Theory	density, functional, theory, electronic, structure	Density Functional Theory	1.00	Electronic Structure Theory	0.52	Electronic Structure and Density	0.65	Electronic Structure Theory	0.52
Integrated Circuit Design	integrated, circuit, design, layout, verification	Integrated Circuit Design	1.00	Integrated Circuit Design	1.00	Integrated Circuit Design	1.00	Integrated Circuit Design	1.00
Image and Video Processing	image, video, processing, enhancement, segmentation	Image and Video Processing	1.00	Image Processing and Enhancement	0.73	Image Processing and Segmentation	0.71	Image Processing Techniques	0.74
Data Visualization & Reporting	visualization, dashboard, reporting, charts, insight	Data Visualization and Reporting	0.99	Data Visualization & Analytics	0.81	Business Intelligence Dashboards	0.56	Data Visualization and Reporting	0.99
Molecular Dynamics Simulations	molecular, dynamics, simulation, atomistic, interactions	Molecular Dynamics Simulation	0.98	Molecular Dynamics Simulation	0.98	Molecular Dynamics and Simulation	0.96	Molecular Dynamics Simulation	0.98
Social Engineering & Phishing	phishing, social, engineering, user, awareness	Phishing and Social Engineering	0.97	Phishing and Social Engineering	0.97	Social Engineering and Phishing	0.98	Phishing and Social Engineering	0.97
Sentiment and Emotion Analysis	emotion, sentiment, opinion, classification, analysis	Emotion and Sentiment Analysis	0.97	Emotional Intelligence and Classification	0.44	Emotional and Sentiment Analysis	0.92	Emotion and Sentiment Analysis	0.97
Complex System Architecture	systems, complex, interoperability, architecture, platforms	Complex Systems Architecture	0.96	Complex Systems	0.65	Complex Systems Architectures	0.92	Complex Systems Architectures	0.92
Statistical Analysis and Inference	statistics, inference, testing, probability, sampling	Statistics and Inference	0.90	Probability and Statistics	0.72	Statistical Inference and Testing	0.81	Statistical Inference Techniques	0.80
Industrial Cybersecurity Threats	cybersecurity, threats, industrial, networks, protection	Industrial Cybersecurity	0.89	Industrial Cybersecurity	0.89	Industrial Cybersecurity and Threats	0.97	Industrial Cyber Security	0.85
ETL Pipeline Design	etl, pipeline, ingestion, transformation, loading	ETL and Pipeline	0.88	ETL Pipelines	0.91	etl-based pipelines	0.84	ETL Pipeline Integration	0.85
Medical & Biological Imaging	medical, imaging, diagnosis, radiology, biometrics	Medical Imaging and Diagnostics	0.88	Medical Imaging and Diagnostics	0.88	Medical Imaging and Diagnostics	0.88	Medical Imaging Techniques	0.84
Low Power Design Optimization	power, low, optimization, analysis, clock	Low Power Optimization	0.88	Low-Power Optimization	0.84	Low Power Optimization	0.88	Low Power Optimization	0.88
Digital Logic Circuit Design	gate, circuit, digital, design, logic	Digital Circuit Design	0.88	Digital Circuit Design	0.88	Digital Circuit Design	0.88	Digital Circuit Design	0.88
Computer Networks and Protocols	network, protocol, wireless, routing, internet	Networking and Protocols	0.87	Wireless Network Design	0.42	Wireless Networking	0.45	Wireless Internet Protocols	0.58

Remote Sensing Technologies	remote, imaging, lidar, radar, satellite	Remote Sensing and Imaging	0.86	Remote Sensing Techniques	0.91	Remote Sensing and Imaging	0.86	Remote Sensing Techniques	0.91
Data Mining and Pattern Discovery	data, mining, extraction, discovery, large	Data Mining and Discovery	0.86	Data Mining Techniques	0.70	Data Mining and Exploration	0.71	Data Mining Techniques	0.70
Predictive Maintenance and Health	prediction, maintenance, failure, health, prognostics	Predictive Maintenance and Failure	0.85	Prognostic Model Prediction	0.66	Health Prognostics and Maintenance	0.82	Maintenance Health Prognosis	0.76
Chemical and Gas Sensing	chemical, gas, detection, surface, selectivity	Chemical Gas Detection	0.85	Chemical Surface Detection	0.61	Chemical Gas Detection	0.85	Chemical Gas Detection	0.85
Nanoelectronic Device Applications	sensors, transistor, nanodevices, nanoelectronics, memristors	Nanoelectronics and Sensors	0.84	Nanomaterial-Based Sensor Devices	0.62	Nanoelectronics and Sensors	0.84	Nanoelectronics and Sensors	0.84
Noise Reduction and Estimation	noise, denoising, estimation, kalman, wiener	Noise and Noise Estimation	0.84	Noise Estimation Techniques	0.83	Noise Estimation Techniques	0.83	Noise Estimation Techniques	0.83
Syntactic Parsing and Grammar	parsing, syntactic, dependency, tree, grammar	Syntactic Parsing Techniques	0.84	Syntactic Parsing and Logic	0.88	Syntactic Grammar and Parsing	0.98	Syntactic Grammar Parsing	0.94
Computer Hardware Architecture	processor, memory, cache, bus, architecture	Processor Architecture and Design	0.83	Processor Architecture Design	0.80	Embedded Systems	0.57	Processor Architectures	0.83
Dialogue Systems and Chatbots	dialogue, chatbot, generation, interaction, system	Chatbots and Interaction	0.82	Chatbots and Interaction	0.82	Artificial Intelligence Systems	0.49	Chatbot Generation and Interaction	0.75
Circuit Simulation Techniques	simulation, circuit, spice, transistor, modeling	Circuit Simulation and Modeling	0.82	Circuit Simulation and Modeling	0.82	Circuit Simulation and Simulation	0.85	Circuit Simulation Techniques	1.00
Big Data Analytics	big, data, analytics, real-time, optimization	Big Data and Analytics	0.82	Big Data Analytics	1.00	Big Data Analytics	1.00	Big Data Analytics	1.00
Targeted Drug Delivery Systems	drug, delivery, targeted, therapeutics, nanocarrier	Drug Delivery Systems	0.81	Targeted Drug Delivery	0.92	Targeted Drug Delivery	0.92	Drug Delivery Systems	0.81
Boundary Detection and Refinement	boundary, edge, contour, detection, precision	Boundary and Edge Detection	0.79	Boundary and Edge Detection	0.79	Boundary Edge Detection	0.79	Boundary Edge Detection	0.79
Reinforcement Learning	reinforcement, agent, environment, reward, policy	Reinforcement Learning Theory	0.79	Reinforcement Learning Agents	0.76	Reinforcement Learning Agents	0.76	Reward and Environment	0.41
Image Recognition and Vision	image, vision, segmentation, recognition, detection	Image Recognition and Segmentation	0.79	Image Recognition and Segmentation	0.79	Vision and Image Recognition	0.93	Image Processing Techniques	0.57
Data Visualization & Insight	visualization, dashboard, charts, insight, storytelling	Data Visualization and Analytics	0.79	Data Visualization & Analytics	0.80	Visualization and Storytelling	0.52	Data Visualization Techniques	0.72
Image and Video Segmentation	segmentation, pixels, regions, clustering, boundary	Image Segmentation and Clustering	0.78	Image Segmentation and Clustering	0.78	Segmentation and Clustering	0.66	Pixel Segmentation Techniques	0.68
Operational Amplifier Design	amplifier, opamp, feedback, gain, bandwidth	Amplifier Design and Analysis	0.77	Amplifier Feedback and Gain	0.51	Amplifiers and Feedback	0.54	Opamp and Feedback Systems	0.58
Video Compression Techniques	compression, encoding, decoding, storage, bandwidth	Video Compression and Storage	0.77	Data Compression Techniques	0.78	Compression and Decoding	0.65	Data Compression and Decoding	0.60
Video Motion Estimation	motion, estimation, flow, analysis, video	Video Motion Analysis	0.77	Motion Estimation Techniques	0.73	Motion Estimation Techniques	0.73	Motion Estimation Techniques	0.73
Signal Compression and Coding	compression, coding, source, lossy, lossless	Video Compression and Coding	0.76	Source Code Coding	0.35	Lossless Compression Techniques	0.72	Lossless Compression Techniques	0.72
Nanomaterials for Biomedical Use	nanomedicine, delivery, targeted, biomedical, biosensing	Nanomedicine and Biomedical Engineering	0.76	Nanomaterial-Based Drug Delivery	0.72	Targeted Biomedical Delivery	0.45	Targeted Biomedical Delivery	0.45
AI Predictive Maintenance	ai, machine, learning, predictive, maintenance	Machine Learning and Maintenance	0.75	AI-Driven Maintenance	0.89	AI-Driven Automation	0.54	Artificial Intelligence and Maintenance	0.85
Generative Image Synthesis	generative, models, synthesis, gans, diffusion	Generative Modeling and Synthesis	0.75	Generative Algorithms	0.61	Generative Neural Networks	0.61	Generative Modelling Techniques	0.60

Smart Factory Automation	smart, factory, manufacturing, automation, systems	Smart Manufacturing Systems	0.75	Smart Factory Automation	1.00	Smart Factory Automation	1.00	Smart Manufacturing Systems	0.75
Statistical Inference	statistics, inference, hypothesis, test, probability	Statistics and Hypothesis Testing	0.75	Probability and Statistics	0.70	Probability and Statistics	0.70	Statistical Inference Methods	0.85
Design for Testability	testing, testability, fault, design, coverage	Software Testing and Design	0.74	Testability and Design	0.93	Testability and Design	0.93	Testability and Design	0.93
Quantum Dot Optoelectronics	quantum, dots, nanocrystals, photoluminescence, semiconductor	Quantum Dots and Photoluminescence	0.74	Quantum Dots and Photoluminescence	0.74	Nanomaterials and Photoluminescence	0.55	Quantum Optics and Nanocrystals	0.72
Autonomous Closed-Loop Control	control, closed-loop, feedback, autonomous, management	Autonomous Control Systems	0.74	Open-Loop Control	0.62	Closed-Loop Control	0.84	Autonomous Control Systems	0.74
Biosensing and Bioelectronics	microfluidics, biological, biomarker, detection, immunosensor	Biosensors and Microfluidics	0.74	Microfluidic Biomarker Sensing	0.48	Microfluidics and Biomarkers	0.40	Microfluidics Biomarkers	0.37
Fog-Cloud Continuum Management	fog, cloud, hierarchy, interoperability, management	Fog and Cloud Management	0.74	Fog and Cloud Management	0.74	Cloud Management Systems	0.47	Cloud Management Systems	0.47
Malware Analysis and Detection	malware, analysis, reverse, engineering, sandbox	Malware Analysis and Reverse Engineering	0.74	Reverse Engineering Techniques	0.40	Malware Analysis and Engineering	0.71	Reverse Engineering Techniques	0.40
Explainable AI (XAI)	explainable, xai, interpretability, transparency, feature	Xai and Interpretability	0.74	Explainable AI	0.89	Explainable Information Architectures	0.71	Xai and Interpretability	0.74
Embedded Systems and Firmware	embedded, system, firmware, microcontroller, realtime	Embedded Systems Design	0.72	Embedded System Design	0.72	Embedded Systems & Firmware	0.99	Embedded Systems Design	0.72
Deep Learning Models	neural, deep, learning, network, machine	Deep Neural Networks	0.72	Deep Neural Networks	0.72	Deep Neural Networks	0.72	Deep Learning Techniques	0.72
Deep Learning Models	neural, networks, deep, training, architecture	Deep Neural Networks	0.72	Deep Neural Networks	0.72	Deep Neural Networks	0.72	Deep Learning Architectures	0.75
Secure Access Control	access, control, authentication, authorization, identity	Identity and Access Control	0.71	Identity and Access Control	0.71	Access Control and Identity	0.71	Identity and Access Control	0.71
Edge and Cloud Integration	edge, computing, latency, fog, devices	Edge Computing and Networking	0.70	Edge Computing and Fog	0.61	Edge Computing Devices	0.62	Edge Computing and Devices	0.66
Industrial IoT & Sensors	internet, things, iiot, sensors, connectivity	Internet of Things and Sensors	0.69	IoT Sensor Networks	0.64	IoT and Things	0.54	Internet of Things and Sensors	0.69
Environmental Nanocatalysis	catalysis, nanocatalysts, environmental, sorption, purification	Environmental Catalysis and Purification	0.69	Environmental Catalysis	0.74	Catalysis and Nanomaterials	0.76	Environmental Catalysis Techniques	0.72
IoT Device Integration	iot, sensors, devices, connectivity, gateway	IoT and Sensors	0.69	IoT Device Networking	0.70	IoT Device Management	0.79	IoT Devices and Connectivity	0.71
Image Feature Extraction	feature, extraction, descriptor, matching, retrieval	Feature Extraction and Matching	0.68	Feature Extraction and Matching	0.68	Feature Extraction and Retrieval	0.73	Descriptor Analysis Techniques	0.51
Segmentation Dataset Annotation	annotation, ground, truth, dataset, labeling	Data Annotation and Labeling	0.68	Ground Truth Annotation	0.55	Annotation Methods	0.55	Data Annotation Techniques	0.60
3D Computer Vision	three-dimensional, reconstruction, geometry, pose, depth	Three-Dimensional Reconstruction	0.67	Three-D Model Reconstruction	0.59	Three-Dimensional Reconstruction	0.67	Three-Dimensional Reconstruction	0.67
Nanomaterial-Based Sensor Technology	sensor, detection, chemical, biosensor, sensitivity	Sensor Technology and Sensing	0.67	Chemical Sensor Detection	0.54	Biosensor Sensor Detection	0.54	Biosensors and Sensors	0.66
Deep Learning Architectures	neural, networks, deep, learning, model	Deep Neural Networks	0.67	Deep Learning and Neural Networks	0.69	Deep Neural Networks	0.67	Deep Learning Models	0.75

Analog-to-Digital Conversion	converter, digital, sampling, resolution, noise	Digital Signal Processing	0.67	Digital Signal Processing	0.67	Digital Signal Processing	0.67	Digital Audio Converters	0.71
Analog-to-Digital Conversion	sampling, conversion, analog, digital, quantization	Digital Signal Processing	0.67	Analog to Digital Converters	0.92	Analog Signal Processing	0.71	Digital Quantization Techniques	0.49
Digital Filter Design	filter, design, fir, iir, adaptive	Adaptive Filter Design	0.66	Adaptive Filter Design	0.66	Intelligent Filter Design	0.71	Advanced Filter Design	0.80
Intrusion Detection Systems	intrusion, detection, anomaly, network, system	Intrusion Detection and Analysis	0.66	Intrusion Detection Systems	1.00	Intrusion Detection Systems	1.00	Network and System Security	0.55
Analog Device Modeling	transistor, mosfet, bit, device, modeling	Transistor Device Modeling	0.66	Device Modeling	0.71	Embedded Systems	0.25	MoSFET Device Modeling	0.57
Data Quality & Governance	governance, quality, metadata, lineage, compliance	Digital Governance and Quality	0.65	Metadata Governance and Compliance	0.62	Metadata Governance and Compliance	0.62	Metadata Governance and Quality	0.70
IoT Security Protocols	iot, device, protocol, security, firmware	IoT and Device Security	0.65	IoT Device Security	0.68	IoT Device Security	0.68	IoT Device Security	0.68
Data Preprocessing and Quality	cleaning, wrangling, transform, imputation, quality	Data Cleaning and Wrangling	0.65	Transform Quality Assurance	0.50	Cleaning and Transformation Techniques	0.44	Imputational Quality Control	0.52
Nanocomposite and Nanocoating Development	coating, composite, mechanical, reinforcement, polymer	Composite Materials and Coatings	0.64	Composite Coatings	0.65	Composite Materials and Coating	0.62	Polymer Composite Materials	0.46
Business Intelligence and KPIs	business, metrics, kpi, strategy, performance	Business Metrics and Strategy	0.64	Business Performance Measurement	0.59	Business Performance Indicators	0.65	Business Performance Metrics	0.63
Machine Translation Techniques	translation, machine, cross-lingual, parallel, transfer	Cross-Linguistic Translation	0.64	Cross-Lingual Translation	0.62	Translation Machines	0.78	Translation Machines	0.78
Biometric Security Patterns	biometric, face, fingerprint, authentication, identity	Biometric Identity and Authentication	0.64	Biometric Identity Authentication	0.67	Biometric Identity and Authentication	0.64	Biometric Identification Systems	0.63
Security and Trust in Edge	security, intrusion, authentication, protection, trust	Information Security and Trust	0.63	Authentication and Security	0.51	Security and Intrusion Prevention	0.42	Information Security and Protection	0.42
Character and Text Recognition	text, handwriting, character, optical, documents	Text and Handwriting	0.63	Optical Character Recognition	0.77	Text-to-Text Handwriting	0.51	Text and Handwriting	0.63
Carbon Nanomaterials and 2D	graphene, nanotubes, 2d, carbon, heterostructures	Graphene Nanostructures	0.63	Graphene Nanostructures	0.63	Heterostructures in Carbon Nanotubes	0.61	Heterostructures and Nanotubes	0.46
Edge Caching and Content Delivery	caching, content, delivery, proximity, storage	Content Delivery and Storage	0.63	Content Caching and Delivery	0.83	Content Delivery and Delivery	0.64	Content Delivery Techniques	0.56
Modeling Nanomaterials	nanomaterials, graphene, semiconductors, quantum, two-dimensional	Nanomaterials and Devices	0.62	Nanomaterials and Nanostructures	0.67	Nanomaterials and Quantum Materials	0.63	Nanomaterials and Semiconductor Materials	0.60
Big Data Technologies	big, data, processing, scale, storage	Big Data and Storage	0.62	Big Data Analytics	0.73	Big Data Management	0.59	Big Data Processing	0.64
RF and Wireless Circuits	radio, frequency, wireless, microwave, Ina	Radio Frequency Technology	0.62	Wireless Communication Systems	0.65	Wireless Communication Systems	0.65	Wireless Radio and Microwave	0.72
Wireless Sensor Networks	wireless, network, iot, communication, node	Wireless Networking and Communication	0.62	Wireless IoT Networks	0.71	Wireless Networking	0.63	Wireless IoT Communications	0.56
Generative AI Models	generative, adversarial, gan, creation, synthetic	Generative Systems Theory	0.62	Generative Agents	0.73	Generative Artificial Networks	0.75	Generative Design Techniques	0.55
Vision for Robotics	autonomous, driving, robotics, slam, perception	Autonomous Driving and Robotics	0.61	Autonomous Vehicle Perception	0.61	Autonomous Systems in Driving	0.43	Autonomous Vehicle Systems	0.43
Wearable and Flexible Sensors	wearable, implantable, flexible, integrated, monitoring	Wearable Medical Devices	0.61	Wearable Embedded Systems	0.64	Wearable & Implantable Devices	0.56	Wearable Medical Devices	0.61

AI Ethics and Governance	data, privacy, ethics, bias, fairness	Data Ethics and Privacy	0.61	Data Ethics and Privacy	0.61	Data Privacy and Ethics	0.57	Data Ethics and Ethics	0.58
Machine Learning in CE	learning, algorithm, data, neural, intelligence	Machine Learning and Intelligence	0.60	Artificial Intelligence and Learning	0.52	AI for Neural Networks	0.47	Machine Learning and Intelligence	0.60
Computer Vision Techniques	vision, image, recognition, detection, segmentation	Image Processing and Recognition	0.60	Image Recognition & Vision	0.62	Vision and Image Recognition	0.58	Vision and Image Processing	0.63
Data Preprocessing Techniques	data, cleaning, wrangling, prep, imputation	Data Cleaning and Wrangling	0.60	Data Preparation and Cleaning	0.65	Data Processing and Preparation	0.73	Data Cleaning Techniques	0.68
Nanomaterials for Biomedical Imaging	imaging, contrast, biomedical, diagnosis, fluorescence	Biomedical Imaging Techniques	0.59	Biomedical Imaging Techniques	0.59	Biomedical Imaging and Contrast	0.56	Biomedical Imaging Techniques	0.59
Operating System Kernels	operating, kernel, resource, scheduling, system	Operating System Architecture	0.59	Operating Systems	0.64	Operating System Management	0.62	Operating System Scheduling	0.49
First-Principles Material Properties	first-principles, calculation, properties, quantum, material	First-Principles Calculations	0.59	First-Principles Calculations	0.59	First-Principles Calculus	0.43	First-Principles Calculation	0.51
Nanomaterials Synthesis Methods	synthesis, fabrication, chemical, exfoliation, deposition	Chemical Synthesis and Fabrication	0.59	Chemical Synthesis and Fabrication	0.59	Chemical Synthesis and Fabrication	0.59	Chemical Synthesis Techniques	0.66
Text Preprocessing and Embeddings	tokenization, embedding, vocabulary, corpus, pre-processing	Corpus Analysis and Tokenization	0.59	Pre-Processing and Tokenization	0.53	Embedded Systems	0.13	Vocabulary Preprocessing	0.62
Acoustic Speech Pattern Recognition	speech, audio, voice, acoustics, speaker	Speech and Voice Technology	0.58	Speech and Acoustics	0.64	Speech and Acoustics Research	0.63	Speech and Audiology	0.46
Advanced Cryptography Techniques	encryption, decryption, algorithm, cryptography, key	Cryptography and Encryption	0.58	Cryptographic Encryption Algorithms	0.59	Encryption and Cryptography	0.56	Advanced Cryptography Techniques	1.00
Data Warehouse and Lakes	warehouse, lake, schema, modeling, sql	Data Warehouse Modeling	0.58	Data Warehouse Modeling	0.58	Warehouse Modeling & Modelling	0.48	Warehouse Modeling and Modelling	0.48
Sensor Materials and Transducers	piezoelectric, capacitive, memristor, transducer, nanomaterials	Piezoelectric Materials and Devices	0.58	Piezoelectric Transducer Design	0.63	Piezoelectric Transducers and Nanomaterials	0.66	Piezoelectric Nanomaterials	0.39
Explaining Deep Learning	neural, deep, cnn, transformer, visualization	Deep Neural Networks	0.58	Deep Neural Networks	0.58	Deep Neural Networks	0.58	Deep Learning Techniques	0.51
Object Detection and Tracking	detection, tracking, object, recognition, computer	Computer Vision and Object Recognition	0.57	Object Recognition and Tracking	0.92	Object Tracking and Recognition	0.87	Object Recognition Techniques	0.56
Federated Learning at the Edge	federated, learning, privacy, distributed, model	Distributed Learning Models	0.57	Distributed Learning Models	0.57	Distributed Learning and Learning	0.61	Distributed Learning Models	0.57
Deep Learning for Vision	neural, convolutional, learning, deep, networks	Deep Neural Networks	0.57	Deep Neural Networks	0.57	Deep Neural Networks	0.57	Deep Learning Techniques	0.54
Deep Learning for Vision	convolutional, neural, networks, deep, learning	Deep Neural Networks	0.57	Convolutional Neural Networks	0.54	Convolutional Neural Networks	0.54	Convolutional Neuroscience Techniques	0.43
Real-time Edge Data Analytics	realtime, analytics, processing, stream, data	Realtime Analytics and Processing	0.57	Real-Time Data Analytics	0.62	Real-Time Data Processing	0.53	Real-Time Data Processing	0.53
Dislocation Dynamics and Defects	dislocation, dynamics, crystal, plasticity, defects	Crystal Dynamics and Dynamics	0.56	Crystal Dynamics and Dislocation	0.88	Crystallinity and Dislocation	0.75	Crystal Dynamics and Defects	0.77
Segmentation in Medical Imaging	medical, mri, tumor, diagnosis, radiography	Medical Imaging Techniques	0.56	Medical Imaging Techniques	0.56	Medical Imaging Diagnostics	0.58	Medical Imaging Techniques	0.56
Sensing and Signal Processing	array, beamforming, localization, sensors, detection	Beamforming and Sensing	0.56	Beamforming Arrays	0.26	Antenna Arrays and Localization	0.35	Detection Techniques	0.55

Database Systems	nosql, relational, database, tuning, optimization	Database Optimization and Tuning	0.56	NoSQL Database Optimization	0.37	NoSQL Database Optimization	0.37	Database Optimization Techniques	0.58
Kinetic Monte Carlo Modeling	kinetic, monte, carlo, atomistic, events	Kinetic and Atomic Physics	0.54	Kinetic and Atomistic Events	0.49	Kinetic Modeling	0.81	Kinetics and Events	0.50
Cloud Infrastructure Management	virtualization, servers, data, centers, infrastructure	Data Center Infrastructure	0.54	Data Center Virtualization	0.53	Virtualization and Data Centers	0.50	Data Center Infrastructure	0.54
Transformer Language Models	transformer, attention, sequence, language, model	Attention and Language Modeling	0.54	Transformative Modeling	0.56	Transformer Modeling	0.47	Attention Modeling	0.37
Fundamental Pattern Classification	classification, features, algorithm, data, training	Machine Learning and Classification	0.54	Feature Classification and Training	0.54	AI-Based Classification	0.55	Data Classification Techniques	0.53
Image Quality Enhancement	enhancement, filtering, noise, restoration, quality	Audio Enhancement and Restoration	0.53	Noise Reduction Techniques	0.43	Noise Reduction and Enhancement	0.57	Noise Restoration Techniques	0.43
Analog Filter Circuits	filter, active, passive, frequency, signal	Passive Filter Design	0.53	Passive Filtering Techniques	0.52	Active Signal Processing	0.35	Active Noise Filtering	0.41
Core Machine Learning	machine, learning, model, algorithm, prediction	Machine Learning and Modeling	0.53	Machine Learning Algorithms	0.49	Machine Learning and Prediction	0.49	Machine Learning Models	0.54
Analog Power Management	power, supply, regulator, efficiency, management	Power Supply and Management	0.52	Power Supply Optimization	0.53	Energy Supply Management	0.35	Power Supply Management	0.61
Cloud Resource Management	resource, allocation, scheduling, load, balancing	Resource Allocation and Balancing	0.52	Resource Allocation Optimization	0.48	Resource Allocation Optimization	0.48	Resource Allocation and Management	0.58
Predictive Modeling Techniques	regression, classification, clustering, prediction, learning	Regression and Classification	0.52	Regression Learning & Prediction	0.56	Regression and Learning	0.46	Regression Analysis Techniques	0.59
Nano-Energy Storage Devices	storage, battery, supercapacitor, electrode, rechargeable	Rechargeable Battery Technology	0.51	Rechargeable Battery Storage	0.55	Supercapacitors and Battery Storage	0.51	Supercapacitor Energy Storage	0.61
Digital Twin Modeling	digital, twin, modeling, simulation, physical	Digital Modeling and Simulation	0.50	Digital Twin Simulation	0.84	Digital Twin Architectures	0.66	Digital and Twin Simulation	0.79
Nanomaterials for Solar Energy	photovoltaic, solar, energy, efficiency, quantum	Photovoltaic Energy Systems	0.50	Photovoltaic Energy Systems	0.50	Photovoltaic Energy Systems	0.50	Photovoltaic Energy Systems	0.50
Segmentation for Autonomous Systems	autonomous, vehicles, driving, lanes, roads	Autonomous Vehicles and Driving	0.50	Autonomous Vehicle Control	0.50	Autonomous Vehicle Design	0.54	Autonomous Vehicle Systems	0.55
Blockchain Security Applications	blockchain, ledger, decentralized, smart, contract	Blockchain and Decentralized Systems	0.50	Smart Contracts and Blockchain	0.49	Smart Contracts and Blockchain	0.49	Blockchain and Decentralized Systems	0.50
Deep Learning Models for Segmentation	convolutional, unet, mask, fcn, pixel	Convolutional Neural Networks	0.50	Convolutional Neural Networks	0.50	Convolutional Neural Networks	0.50	Convolutional Image Processing	0.52
AI/ML in Cloud	ai, machine, learning, data, analytics	Machine Learning and Analytics	0.50	AI-Driven Data Analytics	0.44	AI and Machine Learning	0.55	Artificial Intelligence and Analytics	0.44
Traditional Segmentation Algorithms	thresholding, clustering, region, watershed, graph	Clustering and Thresholding	0.49	Regional Graph Clustering	0.33	Watershed Clustering Techniques	0.50	Watershed Analysis Techniques	0.40
Data Query and Storage	sql, database, query, storage, warehousing	SQL Database Architecture	0.49	SQL Server Data Warehouse	0.41	SQL Server Architectures	0.30	SQL Database Management	0.54
AR/VR Twin Visualization	augmented, virtual, reality, visualization, interaction	Augmented and Virtual Reality	0.48	Augmented and Virtual Reality	0.48	Virtual Reality and Augmented Reality	0.49	Virtual and Augmented Reality	0.50
Causal and Logical Explanations	causal, counterfactuals, logic, rules, reasoning	Computational Logic and Reasoning	0.48	Causal and Counterfactual Reasoning	0.78	Causal Logic and Reasoning	0.88	Causal Logic and Reasoning	0.88
Cloud-Native Architectures	containerization, kubernetes, docker, orchestration, serverless	Containerization and Orchestration	0.48	Containerization and Orchestration	0.48	Containerization and Orchestration	0.48	Containerization and Orchestration	0.48

Formal Verification Methods	verification, formal, equivalence, assertion, functional	Functional Verification and Equivalence	0.47	Formal Verification Techniques	0.99	Formal Validation Methods	0.89	Formal Verification Techniques	0.99
Big Data Ecosystems	hadoop, spark, distributed, storage, processing	Hadoop and Distributed Processing	0.47	Distributed Data Processing	0.44	Hadoop and Distributed Systems	0.49	Hadoop Data Processing	0.44
Ethical and Regulatory XAI	ethics, fairness, bias, accountability, regulation	Ethical Issues in Medicine	0.46	Ethical Business Practices	0.42	Ethical and Fairness Issues	0.42	Ethics and Fairness	0.38
Green Cloud Computing	green, energy, efficiency, sustainable, power	Green Building and Energy	0.46	Sustainable Energy Efficiency	0.46	Green Energy Efficiency	0.53	Sustainable Construction Practices	0.29
Nanocatalysis and Surface Chemistry	catalysis, reaction, surface, activity, selectivity	Catalysis and Reactions	0.45	Catalytic Reaction Mechanisms	0.38	Catalysis and Selectivity	0.44	Catalysis and Reactions	0.45
Signal Transformation and Analysis	transform, fourier, wavelet, spectral, estimation	Fourier Transform Methods	0.43	Fourier Transform Estimation	0.34	Fourier Transform Methods	0.43	Fourier Methods and Estimation	0.37
Vision Robustness & Security	robustness, adversarial, attacks, generalization, uncertainty	Adversarial Systems and Attacks	0.43	Uncertainty and Adaptation	0.20	Robustness and Uncertainty	0.43	Robust Systems Analysis	0.45
Computation Offloading & Resource Allocation	offloading, computation, latency, mobile, resource	Mobile Computing Architectures	0.42	Mobile Resource Allocation	0.63	Mobile Resource Offloading	0.55	Mobile Resource Management	0.51
Cloud Data Security	security, encryption, privacy, access, authentication	Information Security and Privacy	0.42	Security and Privacy	0.47	Data Security and Privacy	0.60	Data Security and Privacy	0.60
Real-time Cyber-Physical Modeling	modeling, simulation, physical, cyber, real-time	Cyber Security Modeling	0.39	Real-Time Systems Simulation	0.61	Real-Time Computer Modeling	0.70	Physical Modeling and Simulation	0.60
Industrial IoT Optimization	industrial, iot, manufacturing, operations, optimization	Industrial Operational Excellence	0.39	Industrial IoT Optimization	1.00	IoT in Industrial Operations	0.77	Industrial IoT Optimization	1.00
Human-Centered XAI	human, user, interface, evaluation, fidelity	Human-Computer Interaction	0.37	Human-Centered Design	0.53	Human-Computer Interface Evaluation	0.37	Human-Software Interface Evaluation	0.41
Phase Field Microstructure Modeling	phase, field, microstructure, evolution, grain	Microstructure and Evolution	0.36	Microstructure and Field Evolution	0.61	Microstructure and Evolution	0.36	Grain Microstructure and Evolution	0.35
Sensor Characterization and Metrology	calibration, noise, accuracy, drift, linearity	Calibration and Error	0.35	Calibration Noise and Accuracy	0.37	Analog Signal Processing	0.38	Optical Noise Control	0.15
Core Computer Vision Tasks	detection, classification, recognition, segmentation, tracking	Machine Learning and Detection	0.34	Object Recognition and Classification	0.47	Detection and Classification	0.40	Detection Techniques	0.30
Data-Driven Twin Analytics	data, acquisition, sensors, integration, analytics	Data Acquisition and Analytics	0.33	Data Acquisition and Integration	0.23	Data Acquisition and Integration	0.23	Data Acquisition and Integration	0.23
AI in EDA	machine, learning, ai, bayes, reinforcement	Machine Learning and Bayes	0.30	Artificial Intelligence and Machine Learning	0.51	AI for Machine Learning	0.57	Machine Learning Techniques	0.34
Advanced CPU Design	pipeline, risc, isa, cpu, performance	Pipeline and System Performance	0.30	Isa and Pipeline Performance	0.34	IoT Pipelines and Performance	0.22	Cpu Performance Analysis	0.58
Explanation Components	features, relevance, importance, attribution, saliency	Features and Relevance	0.30	Relevance and Feature Attribution	0.29	Relevance and Saliency	0.29	Relevance and Attribution	0.31
Workflow Orchestration	monitoring, automation, airflow, orchestration, scheduling	Airflow Control and Monitoring	0.29	Controlled Airflow Optimization	0.20	Automated Airflow Management	0.36	Airflow Control Systems	0.26
Perovskite Nanocrystal Applications	perovskite, solar, cells, optoelectronic, nanocrystals	Optoelectronic Devices	0.26	Perovskite Solar Cells	0.78	Perovskite Solar Cells	0.78	Advanced Optoelectronic Materials	0.35
Foundations of XAI	interpretability, transparency, black-box, trust, auditability	Transparency and Interpretability	0.25	Transparency and Trust	0.09	Transparency and Interpretability	0.25	Accountability and Transparency	0.07
Types of Image Segmentation Tasks	semantic, instance, panoptic, classes, delineation	Semantics and Instances	0.15	Semantic Class Delineation	0.29	Semantic Class Delineation	0.29	Semantic Class Delineation	0.29

Physical Design Automation	place, route, layout, geometry, physical	Place and Geometry	0.14	Physical Modeling	0.40	Physical Location and Layout	0.31	Physical Place and Geometry	0.24
Post-Hoc Explanation Methods	lime, shap, surrogate, local, agnostic	Limelight and Surrogacy	0.02	Surrogate Networks	0.25	Agnostic Lime Modeling	0.08	Lime Shape Analysis	0.03

	ver_base_sim	tax_base_sim	ver_large_sim	tax_large_sim
mean	0.627598	0.626594	0.644225	0.645342
std	0.200882	0.206089	0.196983	0.194867
min	0.081399	0.030687	0.085026	0.017294
max	1.000000	1.000000	1.000000	1.000000